

LESSON 2-5 • SESSION 2 • PRACTICE • FORM A

Comparing Variability

Learning target – I can compare the variability of two data sets using range and IQR. · 6.SP.B.5c

REMEMBER

- **More consistent** = less spread out = smaller range or smaller IQR.
- Compare the **same** measure for both groups. Equal ranges can hide different IQRs.
- With one far-out value, trust the **IQR** more than the range.

WORKED EXAMPLE

Class R: range 18, IQR 12. Class S: range 35, IQR 25.

Step 1 Ranges: $18 < 35$.**Step 2** IQRs: $12 < 25$.**Step 3** Both measures are smaller for R.**Answer** **Class R is more consistent****WORDS** **consistent** (*constante*) · **range** (*rango*) · **IQR** (*rango intercuartil*)**PRACTICE***show your work · label units*

- 1. Two Classes.** Class R scores: 70, 82, 76, 88, 73, 85, 79. Class S scores: 60, 90, 75, 95, 65, 80, 72. Complete the table.

Hint Order each list on its own first.

	Range	Q1	Q3	IQR
Class R				
Class S				

More consistent _____

- 2. Vending Machines.** Machine 1 daily sales: 30, 42, 35, 50, 38, 46, 33. Machine 2: 34, 39, 41, 37, 43, 40, 45.
- a.** Range 1: ____ **b.** IQR 1: ____ **c.** Range 2: ____ **d.** IQR 2: ____

- 3. Devi's Claim.** Sets V and W both have a range of 40. V has $Q1 = 15$ and $Q3 = 35$; W has $Q1 = 25$ and $Q3 = 30$. Devi says they vary the same amount.

Hint Equal ranges only match the extremes.☐ Right☐ Not right**IQRs** _____

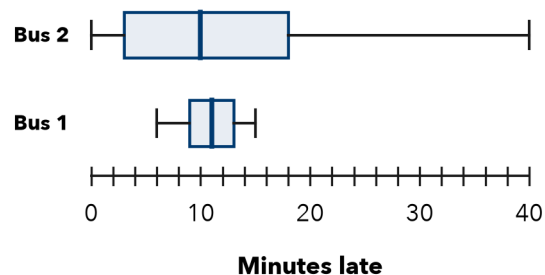
- 4. Hugo's Claim.** Sets T and U both have an IQR of 12. T runs from 30 to 54; U runs from 8 to 96. Hugo says they are equally spread out.

Hint Equal IQRs only match the middle halves.☐ Right☐ Not right**Ranges** _____

- 5. Choosing a Measure.** A list of home prices has one house priced far above all the others. Which measure of spread fits better, the range or the IQR? Explain.

- 6. Different Stories.** Set M has range 40 and IQR 6. Set N has range 24 and IQR 18.
Hint A range much bigger than the IQR points to an extreme value.
- a. Wider middle half: ____ b. Likely has one unusual value: ____
- 7. Phone Batteries.** Phone 1 hours: 16, 20, 18, 24, 22, 26, 14. Phone 2 hours: 19, 21, 20, 23, 18, 24, 17.
- a. Range 1: ____ b. IQR 1: ____ c. Range 2: ____ d. IQR 2: ____

- 8. Choosing a Bus.** The box plots show how many minutes late each bus was.
- a. IQR of Bus 1: ____
- b. IQR of Bus 2: ____
- c. More predictable: ____
- d. Bus 2 is sometimes almost on time. Why might you still choose Bus 1?



FIND THE ERROR *explain the mistake, then fix it*

- 9. Harlan's Work.** Harlan wrote: "Set A has the greater IQR, so Set A must have the greater range too." Explain the mistake.

TEST PRACTICE *choose the best answer*

- 10.** A smaller IQR means the middle half of the data is –
A more spread out **B** closer together **C** greater **D** missing values
- 11.** A range much larger than the IQR suggests –
A an extreme value **B** no variability **C** equal quartiles **D** a small data set

SELF-CHECK CODE *find your answer · copy its letter into that problem's box · spell a math word*

5 I	8 Y	15 G	Class S Q	13 A	4 N	17 Z	Class R V	9 X	6 R
#1		#2b	#2d	#7b	#7d	#8a	#8b		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____